Module 6 — Organic Chemistry and Analysis

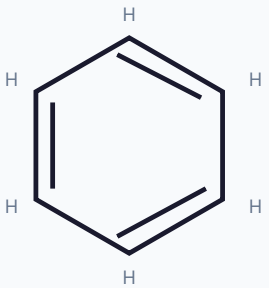
6.1 Aromatic Chemistry

Structure of Benzene

Benzene (C_6H_6): A planar, cyclic molecule with six carbon atoms in a regular hexagonal ring. Each carbon is sp^2 hybridised, forming three sigma bonds (to two adjacent carbons and one hydrogen) at 120°. The remaining unhybridised p orbital on each carbon overlaps sideways with its neighbours to form a **delocalised pi system** — a ring of electron density above and below the plane. This delocalisation makes benzene significantly more stable than the hypothetical "cyclohexatriene" structure proposed by Kekule.

Benzene: Kekule vs Delocalised Model

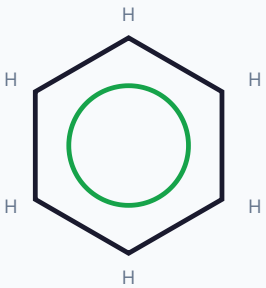
Kekule Model
(cyclohexatriene — INCORRECT)



Problems with this model:
- Predicts alternating bond lengths
- Predicts addition reactions (like alkenes)
- Hydrogenation enthalpy too exothermic (3× cyclohexene value expected)

vs

Delocalised Model
(correct modern model)



Evidence for this model:
- All C-C bonds equal (0.139 nm)
- Undergoes substitution not addition
- More stable by ~150 kJ mol^{-1} (delocalisation/resonance energy)

Evidence against the Kekule model and for delocalisation:

- **Bond lengths:** All C–C bonds in benzene are identical (0.139 nm), intermediate between a C–C single bond (0.154 nm) and a C=C double bond (0.134 nm). Kekule's model would predict alternating bond lengths.
- **Enthalpy of hydrogenation:** Cyclohexene (one C=C) has $\Delta_{hyd}H = -120$ kJ mol^{-1}. If benzene were cyclohexatriene (three C=C), we'd expect $3 \times -120 = -360$ kJ mol^{-1}. The experimental value is only -208 kJ mol^{-1}, meaning benzene is **152 kJ mol^{-1} more stable** than expected. This extra stability is the **delocalisation (resonance) energy**.
- **Resistance to addition:** Benzene undergoes substitution rather than addition reactions, preserving the stable delocalised ring system. True alkenes readily undergo addition reactions.

Electrophilic Substitution of Benzene

> **Electrophilic substitution:** A reaction in which an electrophile attacks the delocalised pi electron system of the benzene ring, substituting for a hydrogen atom. The delocalised ring is temporarily disrupted but is **restored** in the final product, maintaining the thermodynamic stability of the aromatic system.

Nitration

Reagents: concentrated nitric acid + concentrated sulfuric acid (nitrating mixture).
Temperature: below 55 °C (to avoid di- and tri-nitration).
Electrophile: the nitronium ion (NO_2^+), generated by:

$$HNO_3 + H_2SO_4 \rightarrow NO_2^+ + HSO_4^- + H_2O$$

Mechanism: The NO_2^+ electrophile attacks the delocalised pi system, forming a positively charged intermediate (an arenium ion or Wheland intermediate). A proton (H^+) is then lost and the delocalised system is restored. The H_2SO_4 is regenerated — it acts as a catalyst.

Product: Nitrobenzene ($C_6H_5NO_2$), a yellow oily liquid.

Friedel-Crafts Acylation

Reagents: an acyl chloride (e.g. CH_3COCl) with an **$AlCl_3$ catalyst** (a Lewis acid / halogen carrier).

Electrophile: the acylium ion (CH_3CO^+), generated when $AlCl_3$ accepts a lone pair from the chlorine:

$$CH_3COCl + AlCl_3 \rightarrow CH_3CO^+ + AlCl_4^-$$

The acylium ion attacks the benzene ring via electrophilic substitution. The product is an **aromatic ketone** (e.g. phenylethanone, $C_6H_5COCH_3$). $AlCl_3$ is regenerated at the end when H^+ reacts with $AlCl_4^-$ to reform $AlCl_3$ + HCl.

Friedel-Crafts Alkylation

Reagents: a halogenoalkane (e.g. CH_3Cl) with an **$AlCl_3$ catalyst**.

Electrophile: a carbocation (CH_3^+) formed when $AlCl_3$ polarises the C–Cl bond sufficiently.

Product: an alkylarene (e.g. methylbenzene/toluene).

Exam Tip: In Friedel-Crafts mechanism questions, you must show: (1) the role of $AlCl_3$ in generating the electrophile, (2) the curly arrow from the benzene ring to the electrophile, (3) loss of H^+ to restore the ring, and (4) regeneration of the $AlCl_3$ catalyst. $AlCl_3$ is described as a halogen carrier or Lewis acid catalyst.

6.2 Nitrogen Compounds

Amines

Amine: An organic compound derived from ammonia (NH_3) by replacing one or more hydrogen atoms with alkyl or aryl groups. **Primary** amines have one alkyl group (RNH_2), **secondary** have two (R_2NH), and **tertiary** have three (R_3N). **Quaternary ammonium ions** (R_4N^+) have four organic groups.

Preparation of primary amines:

- **From haloalkanes:** React with excess concentrated ammonia in ethanol, heated in a sealed tube. Excess NH_3 ensures the primary amine is the main product (minimises further substitution to secondary, tertiary amines and quaternary ammonium salt).
- **From nitriles:** Reduction with $LiAlH_4$ in dry ether (strong reducing agent), or catalytic hydrogenation with H_2/Ni catalyst. E.g. $CH_3CN \rightarrow CH_3CH_2NH_2$.
- **From nitrobenzene (aromatic amines):** Reduction with tin (Sn) and concentrated HCl, then NaOH to liberate the free amine (phenylamine/aniline).

Amines act as **bases** because the lone pair on nitrogen can accept a proton:

$$CH_3NH_2 + H_2O \rightleftharpoons CH_3NH_3^+ + OH^-$$

Amines also act as **nucleophiles** because the lone pair on nitrogen can attack electrophilic carbon atoms (e.g. in haloalkanes, acyl chlorides).

Basicity order: Alkylamines (e.g. CH_3NH_2) are stronger bases than ammonia because alkyl groups are electron-donating (inductive effect), increasing the electron density on nitrogen and making the lone pair more available. Phenylamine ($C_6H_5NH_2$) is a much weaker base than ammonia because the lone pair on nitrogen is delocalised into the benzene ring, making it less available to accept a proton.

Amino Acids and Chirality

Amino acid: A molecule containing both an amine group ($–NH_2$) and a carboxylic acid group (–COOH). In alpha-amino acids, both groups are attached to the same carbon atom (the alpha carbon). At neutral pH, amino acids exist as **zwitterions** (internal salts) with a positive NH_3^+ group and a negative COO^- group.

Chiral centre: A carbon atom bonded to four different groups. Molecules with a chiral centre exist as two non-superimposable mirror images called **enantiomers**. Enantiomers rotate plane-polarised light in opposite directions (one clockwise, one anticlockwise, by equal amounts). A **racemic mixture (racemate)** contains equal amounts of both enantiomers and shows no net optical rotation — the rotations cancel out.

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All alpha-amino acids (except glycine, where R = H) have a chiral centre at the alpha carbon because it is bonded to four different groups: $–NH_2$, –COOH, –H, and –R.

Exam Tip: Nucleophilic addition of HCN to an aldehyde creates a new chiral centre, producing a racemic mixture. This is because the planar C=O group can be attacked by CN^- from either side with equal probability, forming equal amounts of both enantiomers. Draw 3D representations using wedge (coming towards you) and dash (going away) bonds to show chirality clearly.

Amides

Amides contain the **$–CONH_2$** functional group (primary amide) or –CONHR (secondary) or $–CONR_2$ (tertiary). They are formed by:

- Reaction of a carboxylic acid with an amine under strong heating (slow, equilibrium)
- Reaction of an **acyl chloride with an amine** at room temperature (fast, vigorous, goes to completion — preferred route)

6.3 Polymers

Condensation Polymers

Condensation polymerisation: Monomers with two functional groups join together with the elimination of a small molecule (usually water or HCl) at each link. Unlike addition polymerisation, a by-product is formed. The monomers must be **bifunctional** (have a reactive group at each end).

Polyesters

Formed from a **diol** (two –OH groups) and a **dicarboxylic acid** (two –COOH groups). An ester linkage (–COO–) forms at each join, with water eliminated:

$$\mathrm{nHO–R–OH + nHOOC–R'–COOH \rightarrow (–O–R–OOC–R'–CO–)_n + 2nH_2O}$$

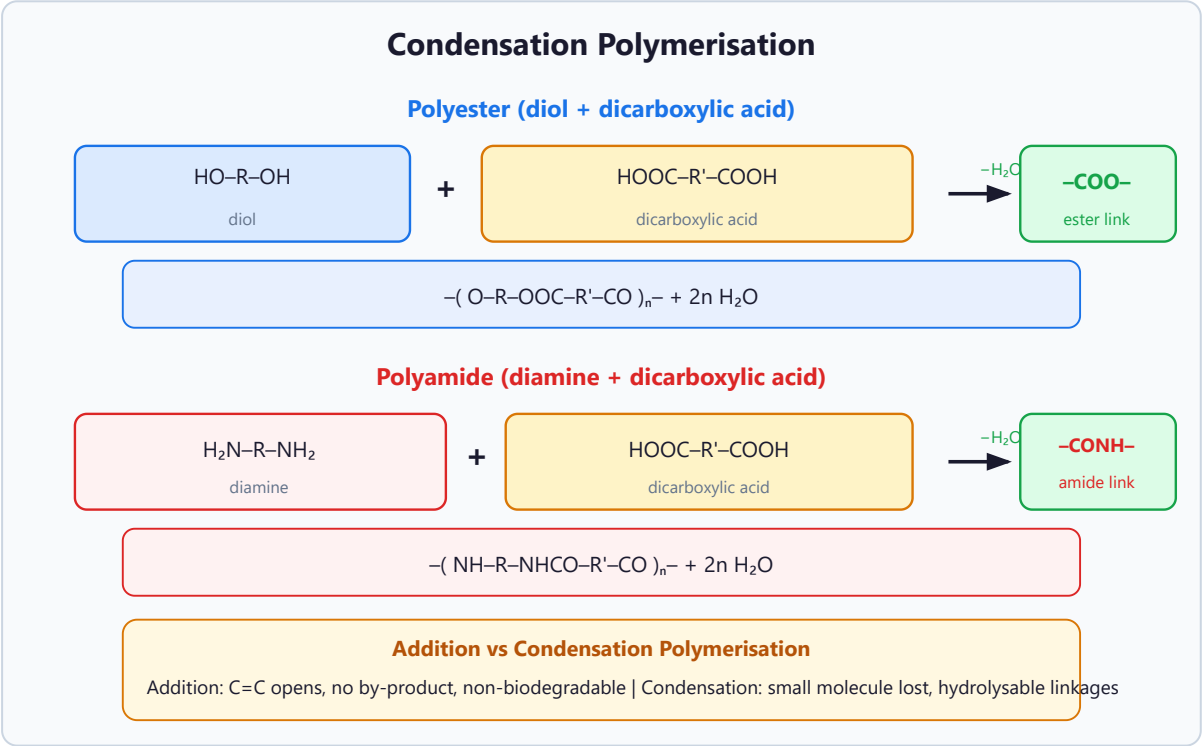
Example: PET (polyethylene terephthalate) is made from ethane-1,2-diol and benzene-1,4-dicarboxylic acid (terephthalic acid). PET is used for plastic bottles, polyester clothing fibres, and food packaging.

Polyamides

Formed from a **diamine** (two $–NH_2$ groups) and a **dicarboxylic acid**. An amide (peptide) linkage (–CONH–) forms at each join:

$$nH_2N–R–NH_2 + nHOOC–R'–COOH \rightarrow (–NH–R–NHCO–R'–CO–)_n + 2nH_2O$$

Examples: Nylon-6,6 (from 1,6-diaminohexane and hexanedioic acid) — used for textiles, carpets, and engineering plastics. Kevlar (from 1,4-diaminobenzene and benzene-1,4-dicarboxylic acid) — used for bulletproof vests and high-strength composites. **Proteins** are natural polyamides made from amino acids joined by peptide bonds.



Biodegradability and Disposal

- **Addition polymers** (e.g. poly(ethene), PVC) are non-biodegradable because the strong C–C backbone is not broken down by microorganisms or hydrolysis. They persist in landfill and oceans for hundreds of years.
- **Condensation polymers** (polyesters, polyamides) can be **hydrolysed** (broken down by water, especially in acidic or alkaline conditions) because the ester or amide linkages are susceptible to hydrolysis. This makes them more biodegradable than addition polymers, though degradation rates vary significantly.

- **Biodegradable polymers** such as PLA (polylactic acid, made from corn starch) and PHB (polyhydroxybutyrate, made by bacteria) can be composted under industrial conditions. They represent a more sustainable alternative to traditional plastics.

Exam Tip: Be able to identify the monomer(s) from a section of a polymer chain and vice versa. For condensation polymers, remember to **add back the water molecule** that was lost when identifying the monomers from a polymer. Look for the ester link (–COO–) in polyesters and the amide link (–CONH–) in polyamides.

6.4 Organic Synthesis and Retrosynthesis

Retrosynthesis: Working backwards from the target molecule to identify suitable starting materials and the reactions needed. You "disconnect" the target at strategic bonds and identify what reagents and conditions would be needed to form those bonds in the forward direction.

Key functional group interconversions:

Conversion	Reagents / Conditions
Alkene → Alcohol	Steam + H_3PO_4 catalyst, 300 °C (industrial hydration)
Alcohol → Alkene	Conc. H_2SO_4 or H_3PO_4, heat (elimination) or Al_2O_3 catalyst
Alcohol → Haloalkane	HBr (or NaBr + H_2SO_4), or PCl_3, PCl_5, or $SOCl_2$ (for chloro)
Haloalkane → Alcohol	Warm aqueous NaOH, reflux (nucleophilic substitution)
Haloalkane → Nitrile (+1C)	KCN in ethanol/water, reflux
Nitrile → Primary amine	$LiAlH_4$ in dry ether (reduction)
Nitrile → Carboxylic acid	Dilute HCl or H_2SO_4, reflux (acid hydrolysis)
Primary alcohol → Aldehyde	$K_2Cr_2O_7$ / H_2SO_4, **distil immediately**

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Primary alcohol → Carboxylic acid	$K_2Cr_2O_7$ / H_2SO_4, **reflux**
Secondary alcohol → Ketone	$K_2Cr_2O_7$ / H_2SO_4, reflux or distil
Carboxylic acid → Acyl chloride	$SOCl_2$ (thionyl chloride) — by-products SO_2 and HCl are gases, easily removed
Acyl chloride + Alcohol → Ester	Room temperature, vigorous reaction (produces HCl fumes)
Acyl chloride + Amine → Amide	Room temperature, vigorous reaction
Aldehyde/Ketone + HCN → Hydroxynitrile	KCN + dilute H_2SO_4 (generates HCN in situ; nucleophilic addition)

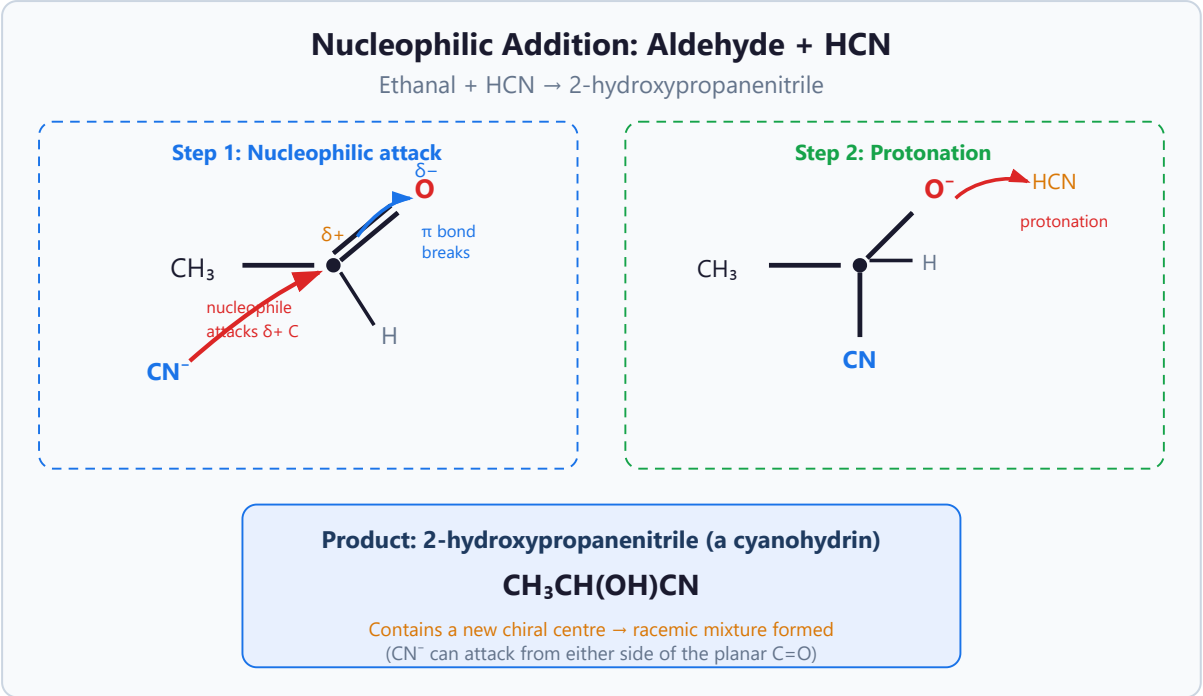
6.5 Carbonyl Compounds

Aldehydes and Ketones

Both contain the C=O (carbonyl) group. Aldehydes have the carbonyl at the **end** of the carbon chain (–CHO); ketones have it in the **middle**. The C=O bond is polar (δ+C=Oδ−) because oxygen is more electronegative than carbon, making the carbon susceptible to nucleophilic attack.

Nucleophilic Addition with HCN

Nucleophilic addition: A nucleophile (CN^-) attacks the electron-deficient (δ+) carbon of the C=O group. The pi bond breaks and the electrons shift to the oxygen, forming an alkoxide intermediate (O^-), which is then protonated by HCN to form a hydroxyl group.



$$CH_3CHO + HCN \rightarrow CH_3CH(OH)CN$$

This reaction is important because it **extends the carbon chain by one** (the CN group adds a carbon) and introduces a chiral centre, producing a racemic mixture. The CN group can subsequently be hydrolysed to –COOH or reduced to $–CH_2NH_2$.

Exam Tip: In practice, KCN + dilute H_2SO_4 is used (generating small amounts of HCN in situ) to avoid handling the highly toxic HCN gas directly. The true nucleophile is CN^-, not HCN. Always show the lone pair on CN^- and the curly arrow from this lone pair to the δ+ carbonyl carbon.

Distinguishing Aldehydes from Ketones

Aldehydes can be oxidised further (they are reducing agents); ketones cannot. Use these tests:

Test	Reagent	Aldehyde Result	Ketone Result
Tollens' reagent (silver mirror test)	$[Ag(NH_3)_2]^+$ (silver nitrate in excess aqueous ammonia)	Silver mirror formed on the tube walls (Ag^+ reduced to Ag)	No visible reaction

Fehling's (or Benedict's) solution	Cu^{2+} complexed in alkaline tartrate solution (blue)	Brick-red precipitate of Cu_2O forms	No visible reaction
$K_2Cr_2O_7$ / H_2SO_4	Acidified dichromate (orange)	Turns green (Cr^{3+}); aldehyde oxidised to carboxylic acid	Stays orange (no reaction)

2,4-Dinitrophenylhydrazine (2,4-DNP or Brady's reagent)

2,4-DNP reacts with **any** carbonyl compound (aldehyde or ketone) to form an orange/yellow precipitate called a **2,4-dinitrophenylhydrazone**. This confirms the presence of a C=O group but does *not* distinguish aldehydes from ketones. The precipitate can be purified by recrystallisation and its melting point compared with data tables to identify the specific compound.

6.6 Carboxylic Acids and Derivatives

Carboxylic Acids

Carboxylic acids contain the –COOH group. They are **weak acids**, dissociating partially in water to release H^+ ions. They undergo typical acid reactions with bases, carbonates, and reactive metals. They have higher boiling points than similar-M_r alcohols because each molecule can form two hydrogen bonds (dimerisation in solution).

Esters

Esters can be prepared by two routes:

- **Fischer esterification:** carboxylic acid + alcohol, refluxed with conc. H_2SO_4 catalyst. Reversible reaction — moderate yield.
- **Acyl chloride route:** acyl chloride + alcohol at room temperature. Vigorous, irreversible reaction — high yield. Produces HCl fumes as by-product.

Esters are hydrolysed (broken back down) by heating with dilute acid or dilute alkali (saponification). Alkaline hydrolysis of a fat or oil with NaOH produces soap (sodium salt of a long-chain carboxylic acid) and glycerol.

Acyl Chlorides

Acyl chloride: A compound with the functional group –COCl. Acyl chlorides are highly reactive because the C=O and C–Cl bonds are both strongly polar, making the carbonyl carbon very electrophilic. They undergo **nucleophilic addition-elimination** reactions readily at room temperature without a catalyst.

Nucleophile	Product	Other Product
Water (H_2O)	Carboxylic acid	HCl (fumes)
Alcohol (ROH)	Ester	HCl
Ammonia (NH_3)	Primary amide	HCl
Primary amine (RNH_2)	N-substituted amide	HCl

Acyl chlorides are preferred over carboxylic acids for synthesis because they react more vigorously, go to completion (not reversible), and give **higher yields**.

6.7 Analytical Techniques

Thin-Layer and Column Chromatography

TLC provides a quick qualitative check of purity and identity by comparing R_f values with standards. Column chromatography separates larger quantities of a mixture for preparative purposes. Both rely on differential affinity for stationary and mobile phases.

Gas Chromatography (GC)

The sample is vaporised and carried by an inert carrier gas (N_2 or He) through a long coiled column coated with a liquid stationary phase. Components separate based on their boiling points and their affinity for the stationary phase. A detector produces a chromatogram showing **retention time** (x-axis) and signal intensity (y-axis). The **number of peaks** indicates the number of components; the **area under each peak** is proportional to the amount of that component.

High-Performance Liquid Chromatography (HPLC)

Uses a liquid mobile phase pumped at high pressure through a column packed with fine solid particles. HPLC is used for non-volatile or thermally unstable compounds that cannot be analysed by GC. UV detection is commonly used.

Exam Tip: GC-MS combines gas chromatography with mass spectrometry. GC separates the components of a mixture (each producing a peak at a characteristic retention time), then each component enters the mass spectrometer for identification via its fragmentation pattern and molecular ion peak. This is extremely powerful for identifying unknown compounds in complex mixtures (e.g. drug testing, forensic analysis, environmental monitoring).

6.8 NMR Spectroscopy

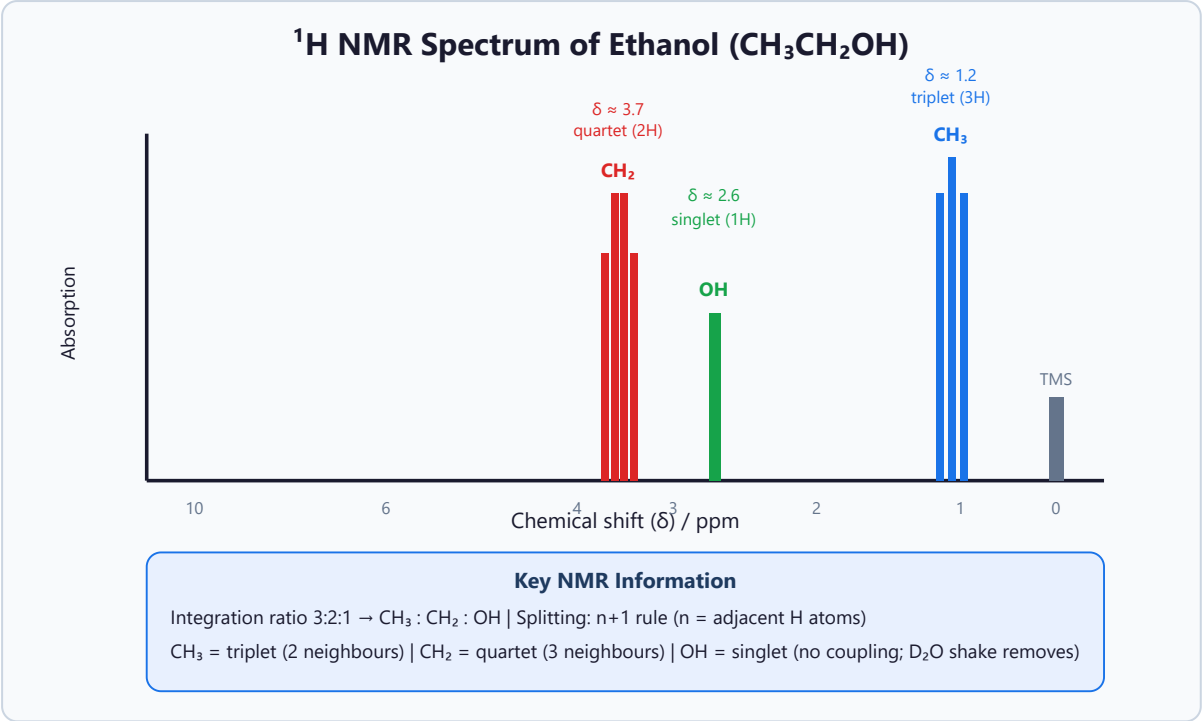
Carbon-13 (^{13}C) NMR

^{13}C NMR: Detects the different chemical environments of carbon atoms in a molecule. Each chemically distinct (non-equivalent) carbon produces a single peak at a characteristic chemical shift (δ, in ppm). The **number of peaks** equals the number of non-equivalent carbon environments. Symmetry reduces the number of peaks (symmetrically equivalent carbons give the same signal).

For example, ethanol (CH_3CH_2OH) has **two** peaks in its ^{13}C NMR: one for the CH_3 carbon and one for the CH_2 carbon (different environments due to proximity to the OH group).

Proton (^{1}H) NMR

^{1}H NMR: Detects the different chemical environments of hydrogen atoms (protons) in a molecule. It provides three key pieces of information:
(1) **Number of peaks** = number of non-equivalent proton environments
(2) **Integration (relative peak area)** = relative number of protons in each environment
(3) **Splitting pattern** = number of non-equivalent protons on adjacent carbon atoms (n+1 rule)



Chemical Shift

The position of a peak on the x-axis, measured in ppm relative to the reference compound **TMS** (tetramethylsilane, $Si(CH_3)_4$), set at δ = 0. TMS is used because: (1) it produces a single sharp peak (all 12 H atoms are equivalent), (2) it is chemically inert and volatile (easily removed), and (3) it absorbs at a higher field (lower δ) than most organic protons, so its peak does not overlap.

Proton Environment	Approximate δ / ppm
R–CH_3 (alkyl)	0.7–1.6
R–CH_2–R (alkyl)	1.2–1.8
R–CH_2–X (near halogen or O)	3.0–4.2
R–O–H (alcohol, variable)	1.0–5.0
R–CHO (aldehyde)	9.0–10.0
R–COOH (carboxylic acid)	10.0–12.0
Ar–H (benzene ring)	6.5–8.0

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Spin-Spin Splitting (n+1 Rule)

A peak is split into **(n + 1)** sub-peaks, where n is the number of non-equivalent protons on adjacent carbon atoms:

Adjacent H Atoms (n)	Pattern	Name	Intensity Ratio (Pascal's triangle)
0	1 peak	Singlet (s)	1
1	2 peaks	Doublet (d)	1:1
2	3 peaks	Triplet (t)	1:2:1
3	4 peaks	Quartet (q)	1:3:3:1

O–H and N–H protons do not usually show splitting and appear as singlets. They can be identified using a **D_2O shake**: adding deuterium oxide causes O–H and N–H peaks to disappear because the labile protons exchange with deuterium (D), which is NMR-silent in ^1H NMR.

Combined Techniques — Strategy

When solving structure determination problems, use all available data systematically:

1. **Molecular formula** from the mass spectrum (M^+ peak) and elemental analysis.
2. **Degree of unsaturation** (Index of Hydrogen Deficiency, IHD): IHD = (2C + 2 + N − H − X) / 2. An IHD of 4 strongly suggests a benzene ring. IHD of 1 suggests one C=C or one C=O or one ring.
3. **Functional groups** from the IR spectrum (look for broad O–H, sharp C=O, etc.).
4. **Carbon environments** from ^{13}C NMR (count distinct peaks; consider molecular symmetry).
5. **Proton environments** from ^1H NMR: count environments, use integration for ratios, use splitting to determine neighbours, use δ values to identify types of protons.
6. **Propose a structure** consistent with ALL the data. Verify by checking every observation matches.

Exam Tip: In combined analytical technique questions, work through the data methodically. Start with M_r from mass spec, calculate IHD, then use IR for functional groups, and finally NMR for the

detailed structure. Cross-reference every piece of data against your proposed structure. The D_2O shake is a common exam point — if a peak disappears, it confirms O–H or N–H.

Worked Example — Structure Determination:

A compound has M_r = 60 (from mass spec M^+ peak). Molecular formula: C_3H_8O.
IHD = (6 + 2 − 8) / 2 = 0 (no rings or double bonds).
IR: broad absorption at 3350 cm^{-1} (O–H), no C=O peak → alcohol, not an aldehyde or acid.
^1H NMR: 4 peaks. δ 0.9 (triplet, 3H), δ 1.6 (sextet, 2H), δ 3.6 (triplet, 2H), δ 2.3 (singlet, 1H — disappears with D_2O).

Structure: propan-1-ol ($CH_3CH_2CH_2OH$). The triplet at δ 0.9 is CH_3 (2 adjacent H); the sextet at δ 1.6 is middle CH_2 (5 adjacent H: 3+2); the triplet at δ 3.6 is CH_2 next to OH (2 adjacent H); the singlet that disappears with D_2O is the OH.